

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

December 2011

IT 206 Object Oriented Programming using Java

Date: 09.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) How do you determine the length of the array? 1
- (b) Group the following under Aggregation or Generalization 2
- a. a-kind-of
 - b. a-part-of
 - c. isa
 - d. or
 - e. and
 - f. part whole
- (c) Differentiate the following 4
1. Event and State
 2. Inheritance and Multiple Inheritance
- Q - 2 (a) Justify the following sentences: 2
- 1.) Qualifications always reduces multiplicity from many to one.
 - 2.) The role name is derived attribute whose value is a set of related objects
- (b) Define following term 6
1. Encapsulation
 2. Polymorphism
 3. Inheritance
- (c) Explain following Keyword 6
1. abstract
 2. protected
 3. final

OR

- Q - 2 (a) Can a class declaration include both the abstract and final modifiers? 2
- (b) What is an applet? Explain the lifecycle of applet. 6
- (c) Why an adapter class is useful? Explain with example. 6
- Q - 3 (a) What are the challenges in designing with inheritance (single/multiple both)? 4
- (b) What are the similarities and differences between character and byte streams? 4
- (c) Declare interface I1 and I2. Interface I3 extends both of these interface. Also declare interface I4. Class X implements I3. Class W extends X and implements I4. Create an object of class W. Use the instanceof operator to test if that object implements each of the interfaces and is of type X. 6

OR

- Q - 3 (a) What is the difference between application and applets? 4
- (b) What is the difference between abstract and concrete classes? 4
- (c) Draw data flow diagram for money transaction process of ATM system 6

SECTION - II

- Q - 4 (a) What is the difference between notify () and notifyAll () methods? 2
- (b) What is the relation between Activity diagram and State diagram? 2
- (c) Explain Constructor overloading with example. 3
- Q - 5 (a) Given these declarations, which of the following expressions are valid? 2

```
byte b = 1;
char c = 1;
short s = 1;
int i = 1;
```

Select the three correct answers.

1. $s = b * 2;$
 2. $i = b << s;$
 3. $b <= s;$
 4. $c = c + b;$
 5. $s += i;$
- (b) When to use the following diagram: 6
1. Package diagram 2. Deployment diagram 3. Interaction diagram
- (c) What is an interface? What are the default modifiers for an interface, interface variable, interface methods? Explain interface inheritance with example. 6

OR

- Q - 5 (a) Why must super () or this () be the first statement in a constructor? 2
- (b) What is run-time polymorphism? Explain dynamic method dispatch. 6
- (c) What is an exception? Name six conditions that generate an exception. Explain how Java searches for a catch block to handle an exception. 6
- Q - 6 (a) Explain following methods with example. 6
1. fillOval () 2. yield () 3. join ()
- (b) Write an application that executes two threads. One thread display "JAVA" every 2,050 milliseconds, and the other displays "INDEPENDENT" every 3,050 milliseconds. Create the thread by implementing the Runnable interface. 8

OR

- Q - 6 (a) Explain following methods with example. 6
1. fillArc () 2. sleep () 3. wait ()
- (b) A customer registers as a member by filling form and mailing it back to the company. No transactions for a period of one-year means he is removed from list and needs to reapply. A member informs the company for any change in personal details. A member makes an order by filling out a sales order form and then faxing it to the company (alternative is by phone). The assistant always checks the validity of the membership. Order processing clerk checks the availability of the ordered items and hold them for the order. If all items are available, clerk schedules delivery. Inventory control clerk controls and maintains the level of stock and responsible for reordering. For queries member phone the assistant. Members may return the defective goods within 30 days and get their money back. Each task carried out by the system has the name and ID of the staff concerned recorded in to the system. 8

1. Draw the activity diagram
2. Draw the sequence diagram